$$W_2 = \frac{15}{16}E^2C \tag{16}$$

$$\frac{l}{v_n} = n\frac{T}{2} \quad n = 1,2,\cdots \tag{1}$$

$$v_n = \frac{2l}{nT} \quad n = 1,2,\cdots \tag{2}$$

$$\Delta t_{n1} = \frac{1}{2}nT \tag{3}$$

$$\Delta t_{n2} = \frac{l}{2v_n} \tag{4}$$

$$t_n = \Delta t_{n1} + \Delta t_{n2} = \frac{3}{4}nT \tag{5}$$

$$t_n \leqslant \frac{10}{4}T \tag{6}$$

$$n_{max} = 3 \tag{7}$$

$$E_n = \frac{1}{2}mv_n^2 + E_0\sin\left[\frac{2\pi}{T}\left(\frac{3n}{4}T\right) + \frac{\pi}{4}\right]$$
$$n = 1,2,3 \tag{8}$$